

A03 Case Study Analysis on Edge-Computing Video Analytics for Real-Time Traffic Monitoring in a Smart City

A. Introduction and Objectives

The Liverpool Smart Pedestrians project aimed to transform an existing city CCTV network into a privacy-compliant, real-time mobility sensing system. Instead of streaming video to a central server or cloud for later analysis, the system performs detection and tracking directly at the camera site and transmits only denatured metadata (counts, trajectories, heatmaps, and time series) to support urban analytics (Barthélemy et al., 2019). The goals were to monitor multiple modes (pedestrians, bicycles, vehicles), reduce latency for “near real-time” insight, and minimize privacy risk by preventing raw video from being stored or transmitted beyond the edge device (Barthélemy et al., 2019).

The project targeted common urban planning challenges: managing congestion and parking pressure in a growing CBD, understanding pedestrian/cyclist safety and movement patterns, and quantifying how redevelopment changes flows over time. Stakeholder workshops were used to align the system outputs with real planning needs (Barthélemy et al., 2019).

B. Methodology

Methodologically, the project followed an iterative, requirements-driven engineering cycle: (1) stakeholder engagement to define requirements, (2) edge device design/prototyping, (3) integration into an IoT platform for interoperability and data sharing, and (4) validation through benchmarking plus real-world deployments (Barthélemy et al., 2019).

Key requirements/constraints shaped sensor development: real-time throughput on embedded hardware; robustness to lighting, viewpoint variation, and crowding; compatibility with existing CCTV feeds; and privacy-by-design constraints (no raw video leaving the device).

Communication options (e.g., Ethernet and LoRaWAN) supported flexible deployment where bandwidth or connectivity is limited (Barthélemy et al., 2019).

C. Technology and Implementation

The smart visual sensor used an NVIDIA Jetson TX2 edge module to run deep learning inference locally, aligning with an edge-computing approach that reduces bandwidth consumption and latency compared to cloud processing (Barthélemy et al., 2019). From a Module 2 perspective, this matches standard edge architecture goals: pushing compute “close to the source” to enable low-latency services and reduce backhaul load (ETSI, 2020).

For detection, the project used YOLOv3 because it offered a strong speed/accuracy balance appropriate for CCTV video where object sizes vary widely with distance (Barthélemy et al., 2019). For tracking, it used SORT, an efficient online tracker (Kalman filter + assignment) designed for real-time multi-object tracking (Barthélemy et al., 2019). Since 2019, real-time detectors have advanced substantially; later YOLO variants (e.g., YOLOv4 and YOLOv7) report improved efficiency/accuracy tradeoffs, which can meaningfully improve edge deployments (Bochkovskiy et al., 2020; Wang et al., 2022).

D. Validation and Performance

Validation was conducted in two stages. First, the authors benchmarked pedestrian detection and tracking on the Oxford Town Centre dataset (1920×1080, 25 fps), using 4,500 frames (≈3 minutes) with 230 annotated pedestrians (Barthélemy et al., 2019). They reported mean accuracy of 0.69 and mean speed of 19.57 fps (median 19.77 fps), demonstrating near-real-time processing on embedded hardware (Barthélemy et al., 2019).

A key performance insight was pipeline bottleneck behavior: YOLOv3 leveraged the GPU, while SORT ran on CPU. As object count increased, tracking became the limiting stage, reflected by an anti-correlation between FPS and detections and periods where GPU utilization dropped when CPU load spiked (Barthélemy et al., 2019). This type of edge bottleneck is consistent with broader findings in edge video analytics—systems often need end-to-end co-optimization (model choice + tracker + acceleration) to sustain real-time traffic monitoring (Wan et al., 2022).

E. Real-World Applications

Indoor deployment (emergency evacuation): The system monitored a stairwell for one hour and produced two peaks aligned with fire alarm evacuation and return-to-building behavior. It detected/tracked 631 people and generated trajectories and heatmaps relevant to safety analytics (Barthélemy et al., 2019).

Outdoor deployment (Liverpool city): The project deployed 20 sensors and generated minute-resolution counts over a week, capturing daily rhythms and reduced Sunday activity. On February 23, 2019, it recorded 20,399 unique objects and produced spatial distributions and trajectory flows that can support decisions such as crosswalk placement, pedestrian–cyclist interactions, and peak loading (Barthélemy et al., 2019).

This aligns directly with Module 3 concepts: the devices operate like **smart sensors + edge gateways**, transforming raw video streams into actionable features (counts/flows) locally and

transmitting only summaries upstream—reducing bandwidth and enabling city-scale rollouts (Barthélemy et al., 2019). This architecture mirrors the broader smart-city trend toward splitting video analytics tasks between edge and higher layers to reduce latency and network load (Pudasaini & Abhari, 2021).

F. Challenges and Future Work

Main challenges included (1) degraded accuracy under occlusion/crowding (undercounting), (2) throughput limits due to CPU-based tracking, and (3) production hardening of the ML pipeline for continuous operation. The authors suggested measuring systematic error for correction factors, offloading tracking to GPU, and moving toward deployment-optimized stacks (e.g., TensorRT-style optimization) (Barthélemy et al., 2019).

New developments since 2019 that could improve similar systems:

- **Edge hardware:** Modern Jetson platforms provide significant increases in AI compute and memory bandwidth compared to earlier generations, making stronger models feasible at the edge (NVIDIA, 2020; NVIDIA, 2023).
- **Detection algorithms:** YOLOv4 and YOLOv7 demonstrate continued improvements in real-time detection performance, often enabling higher accuracy at comparable speed, which directly benefits CCTV-based analytics (Bochkovskiy et al., 2020; Wang et al., 2022).
- **Tracking algorithms:** More robust trackers such as ByteTrack improve association under occlusion by using more detections for identity continuity, which can reduce fragmented trajectories in crowded scenes (Zhang et al., 2022).
- **Edge architecture standards:** Formal MEC reference architectures emphasize low-latency edge app deployment and standardized orchestration concepts, which support more scalable, manageable citywide rollouts (ETSI, 2020).

G. Personal Evaluation

Overall, Liverpool Smart Pedestrians is a strong example of practical smart-city edge AI because it begins with stakeholder-defined requirements, uses existing CCTV infrastructure to reduce deployment cost, and produces planning-relevant outputs while keeping raw video local (Barthélemy et al., 2019). The privacy-by-design approach is a major success: transmitting metadata instead of video reduces privacy exposure and bandwidth, helping scale the solution citywide.

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That said, the reported “accuracy” is best treated as an application-level indicator rather than a complete computer-vision evaluation. For planning, consistent trend capture may matter more than perfect per-frame counts, but future evaluations should include standard metrics (precision/recall by class, occlusion-stratified error, and uncertainty bounds) to interpret small changes confidently. If improving the system today, I would: (1) modernize inference with optimized deployment (e.g., TensorRT) and newer YOLO variants, (2) adopt a more occlusion-robust tracker (e.g., ByteTrack-style association), (3) add self-diagnostics (camera health/drift alerts), and (4) apply privacy threat modeling to reduce re-identification risk from trajectories in sensitive areas.

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